

# CLL — Complications: Infections, Autoimmune Cytopenias & Richter





## 1. Infection deaths

### WHAT YOU SEE

A big '30-50% DEATHS' sign over a patient lying in a hospital bed — infection, not the leukemia itself, is the leading killer.

### WHAT IT ENCODES

Infection is the LEADING cause of death in CLL (~30–50% of deaths), driven by hypogammaglobulinemia, T/NK-cell dysfunction, complement deficiency, neutropenia, and immunosuppressive therapy. Encapsulated bacteria (*S. pneumoniae*, *H. influenzae*), plus opportunistic and viral reactivations.

### BOARD TRAP

Immune compromise in CLL is intrinsic to the disease and present even in untreated, early-stage patients — it is not solely caused by chemotherapy.

## 2. Hypogammaglobulinemia

### WHAT YOU SEE

A shelf of half-empty IgG/IgA/IgM bottles tagged '<300-500 recurrent hospitalized' — falling antibodies that let encapsulated bugs through.

### WHAT IT ENCODES

Hypogammaglobulinemia (low IgG/IgA/IgM) is common and progressive, causing recurrent sinopulmonary infections with encapsulated organisms. Replacement IVIG/SCIG is indicated when IgG is low (typically <500 mg/dL) AND there are recurrent serious/hospitalizing bacterial infections — not for low IgG alone.

### BOARD TRAP

Give IVIG only when low IgG is paired with recurrent serious infections; asymptomatic hypogammaglobulinemia is not by itself an indication.

## 3. Vaccines before tx

### WHAT YOU SEE

A panel of vaccine vials — FLU, COVID, SHINGRIX, RSV, PREVNAR 20 — stamped 'BEFORE THERAPY', with 'LIVE VACCINES' crossed out in red.

### WHAT IT ENCODES

Give inactivated/recombinant vaccines EARLY, ideally BEFORE starting therapy (especially anti-CD20 like rituximab/obinutuzumab, which abolish vaccine responses for months): annual influenza, COVID-19, Shingrix (recombinant zoster), RSV, and pneumococcal (PCV20 or PCV15→PPSV23).

### BOARD TRAP

NO LIVE/attenuated vaccines in CLL (note the crossed-out LIVE VACCINES) — e.g., avoid live zoster (Zostavax), MMR, live nasal flu, yellow fever. Shingrix is recombinant and IS allowed.

## 4. Pneumococcal response

### WHAT YOU SEE

A caption '58% vs 100% controls' beside the pneumococcal vaccine — the blunted seroconversion in CLL versus healthy people.

### WHAT IT ENCODES

Vaccine responses are blunted in CLL (e.g., pneumococcal seroconversion ~58% vs ~100% in healthy controls), worsened by anti-CD20 therapy — reinforcing the rule to vaccinate as early as possible and before B-cell-depleting agents.

## 5. AIHA

### WHAT YOU SEE

A red 'AIHA' curtain over destroyed red blood cells labeled 'warm IgG' — antibody-coated RBCs being torn apart in the periphery.

### WHAT IT ENCODES

Autoimmune hemolytic anemia is the most common autoimmune complication of CLL: WARM-type, IgG-mediated, with positive direct antiglobulin test (DAT/Coombs), high LDH/indirect bilirubin, low haptoglobin, and reticulocytosis.

### BOARD TRAP

AIHA-driven anemia ≠ Rai stage III marrow failure — autoimmune cytopenia does NOT upstage Rai. The anemia is peripheral destruction, not marrow infiltration.

## 6. AIHA treatment

### WHAT YOU SEE

Three first-line boxes — steroids, IVIG, rituximab — in the AIHA room: the autoimmune-directed pathway, separate from treating CLL bulk.

### WHAT IT ENCODES

First-line AIHA in CLL = CORTICOSTEROIDS, with IVIG and/or rituximab added for severe/refractory cases — this is the autoimmune-directed pathway, distinct from treating the CLL bulk. CLL-directed therapy is reserved for refractory autoimmune disease or concurrent treatment-requiring CLL.

### BOARD TRAP

Treat the autoimmune process first with steroids — do NOT jump to CLL chemoimmunotherapy for isolated AIHA/ITP. (Avoid fludarabine, which can trigger/worsen AIHA.)

## 7. Coombs positive

### WHAT YOU SEE

A 'COOMBS / DAT POSITIVE' card on the desk — the test that proves the anemia is immune destruction, not marrow replacement.

### WHAT IT ENCODES

A positive direct antiglobulin test (DAT/Coombs) confirms immune-mediated (warm IgG) hemolysis and distinguishes autoimmune anemia from marrow-replacement anemia — the key test that determines management and staging implications.

## 8. ITP + TPO agonist

### WHAT YOU SEE

A blue 'ITP' curtain with a lone platelet, steroid+IVIG+rituximab boxes, and a 'TPO AGONIST' label for refractory cases.

### WHAT IT ENCODES

Immune thrombocytopenia in CLL = isolated low platelets with normal/increased marrow megakaryocytes; treat first-line with corticosteroids ± IVIG ± rituximab. TPO receptor agonists (eltrombopag/romiplostim) are options for refractory cases.

### BOARD TRAP

Like AIHA, autoimmune ITP does NOT upstage Rai to stage IV — distinguish immune destruction from marrow-failure thrombocytopenia.



## 9. Autoimmune ≠ progression

### WHAT YOU SEE

The banner stretched across the scene 'COMPLICATIONS ≠ JUST PROGRESSION' — autoimmune cytopenias are immune-mediated, not tumor growth.

### WHAT IT ENCODES

The scene's banner: autoimmune cytopenias are IMMUNE-MEDIATED, not a marker of CLL tumor progression, and are managed separately (steroids/rituximab/IVIG) from CLL bulk reduction. Distinguishing autoimmune from marrow-failure cytopenia changes both staging and treatment.

### BOARD TRAP

Do not treat AIHA/ITP as Rai stage III/IV requiring CLL chemoimmunotherapy — that misclassifies prognosis and selects the wrong therapy.

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## 10. Richter transformation

### WHAT YOU SEE

A small CLL cell with an arrow morphing into a large angry DLBCL cell, tagged '0.5%/yr' — the aggressive transformation, plus an owl 'HL variant'.

### WHAT IT ENCODES

Richter transformation = CLL/SLL evolving into an aggressive large B-cell lymphoma (most commonly DLBCL; rarely Hodgkin variant), incidence ~0.5%/yr (cumulative 2–10%). Higher risk with unmutated IGHV, TP53/NOTCH1 abnormalities, and complex karyotype.

### BOARD TRAP

Sudden, asymmetric, RAPIDLY enlarging single node + B symptoms + markedly elevated LDH + hypercalcemia = suspect Richter; you must BIOPSY it to confirm.

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## 11. PET hot spot

### WHAT YOU SEE

A PET scan glowing with a 'HOT SPOT', tag 'SUV≥10 → biopsy' vs 'SUV<5 → CLL' — PET points to the hottest node to biopsy.

### WHAT IT ENCODES

PET-CT localizes the most metabolically active node to target for EXCISIONAL biopsy when Richter is suspected: SUVmax >5 (especially ≥10) raises suspicion and guides where to biopsy; PET cannot replace biopsy for diagnosis.

### BOARD TRAP

SUVmax <5 makes Richter unlikely (consistent with CLL); SUVmax >5 → biopsy the hottest node. PET directs the biopsy, it does not diagnose Richter by itself.

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## 12. Clonal relation/prognosis

### WHAT YOU SEE

Two hanging tags — a skull 'RELATED → 6-16 mo' versus a star 'UNRELATED → 5 yr' — clonal link to the original CLL sets the prognosis.

### WHAT IT ENCODES

Prognosis hinges on clonal relationship to the original CLL: clonally RELATED Richter (~80% of cases) is chemoresistant with dismal median survival ~6–16 months; clonally UNRELATED Richter behaves like de novo DLBCL with far better outcomes (multi-year survival).

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## 13. 3x skin cancer

### WHAT YOU SEE

A bold '3x' sign with a sun and 'SKIN 8-15x', plus a 'YEARLY EXAM' calendar — sharply raised skin-cancer risk needing yearly dermatologic checks.

### WHAT IT ENCODES

CLL substantially increases second-malignancy risk via immune surveillance failure — skin cancers especially (squamous cell, basal cell, and aggressive Merkel cell), up to 8–15× the general population. Requires diligent sun protection and at least yearly dermatologic skin exams.

### BOARD TRAP

Don't dismiss new/changing skin lesions in CLL — aggressive SCC and Merkel cell carcinoma are markedly overrepresented and can metastasize.

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